Problem 01: Treasure Trees

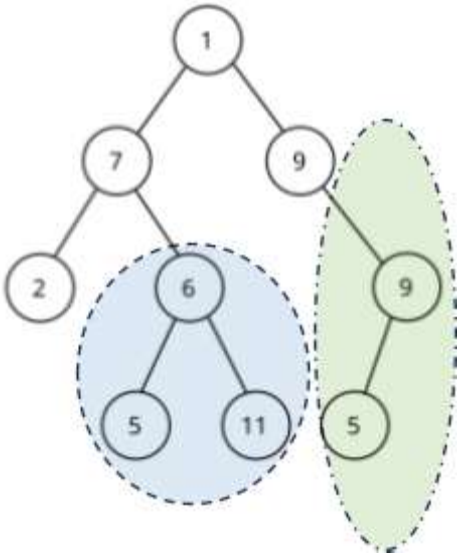
Time limit: 3 seconds

Raju and Rani are exploring a magical forest filled with enchanted trees. Each tree has numbers carved into its nodes, and some of these trees follow a very special rule—they are **Perfect Order Trees** (just like Binary Search Trees).

A **Perfect Order Tree** must obey these rules:

1. Every node on the **left side** must have numbers **strictly smaller** than the number on the parent node.
2. Every node on the **right side** must have numbers **strictly larger** than the parent node.
3. Both left and right sides must themselves also be Perfect Order Trees.

Raju discovers that each subtree hides a **treasure**, with the treasure value equal to the **sum of all numbers** in that subtree.



Rani's challenge to Raju is: *"From this entire magical tree, find the subtree that is a Perfect Order Tree and gives the **maximum treasure value**. Tell me what the highest treasure sum is!"*

So your task is the same as Raju's: **Given the tree, find the maximum sum of any subtree that satisfies the Perfect Order (BST) rules.**

Input

Node values separated by spaces. Positive values showing node values and negative values showing null/ empty child.

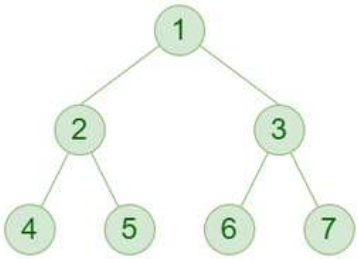
Output

Print value of valid maximum binary search tree.

Input Example 1

Input: 1 7 9 2 6 -1 9 -1 -1 5 11 5
Output: 22
Explanation: There are two binary search trees. The maximum binary search tree has sum 22.

Input Example 2



Input: 1 2 3 4 5 6 7
Output: 7
Explanation: Maximum sum in a valid Binary search tree is 7

Constraints:

The number of nodes in the tree is in the range **1 to 10^5**.
All node values are in range **-10^6 to 10^6**